Option Pricing in Python:
Six Models Checked Against Each Other

Gianni Pilotti

1 The Core Idea

A pricing model is only useful if its output can be trusted. The repository builds six small pricers from scratch (Black-Scholes, binomial tree, Greeks, a barrier option, delta hedging, option strategies) and checks each one against a second method.

All notebooks use the same inputs: spot $S = 100$, strike $K = 100$, rate $r = 3\%$ and volatility $\sigma = 20\%$. Maturity is one year, except for the hedging and strategy notebooks (three months).

- **Closed form against numerics:** the tree converges to Black-Scholes, the Greeks match finite differences, the barrier formula matches Monte Carlo.
- **Model against hedge:** a short call hedged with the model's own delta should end with a P&L close to zero, and the error should shrink with more rebalancing.

2 Method

2.1 Black-Scholes and implied volatility

With $d_1 = \dfrac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$:

$$C = S\,N(d_1) - Ke^{-rT}N(d_2), \qquad P = Ke^{-rT}N(-d_2) - S\,N(-d_1).$$

Put-call parity $C - P = S - Ke^{-rT}$ holds for any volatility and is used as a check. The implied volatility is the root of $C(\sigma) - C_{mkt} = 0$ found with Brent's method on [0.0001, 5].

2.2 Binomial tree (Cox-Ross-Rubinstein)

With n steps and $\Delta t = T/n$:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = 1/u, \quad p = \frac{e^{r\Delta t} - d}{u - d}, \quad V_t = e^{-r\Delta t}\left(pV_{t+1}^{up} + (1-p)V_{t+1}^{down}\right).$$

For an American option, the value at each node is $\max(V_t,\ \text{intrinsic value})$.

2.3 Greeks

Closed forms for the call, with φ the normal density, in the units used by the code:

$$\Delta = N(d_1), \qquad \Gamma = \frac{\varphi(d_1)}{S\sigma\sqrt{T}}, \qquad \mathcal{V} = \frac{S\varphi(d_1)\sqrt{T}}{100},$$
$$\Theta = \frac{1}{365}\left(-\frac{S\varphi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)\right), \qquad \rho = \frac{KTe^{-rT}N(d_2)}{100}.$$

Vega is per vol point, theta per calendar day and rho per 1% of rate. Each is compared with a central finite difference with a bump of 10^{-4}.

2.4 Down-and-out call

For a barrier $H \le K$ watched continuously, the price is the vanilla call minus the down-and-in call. With $\lambda = (r + \frac{1}{2}\sigma^2)/\sigma^2$ and $y = \dfrac{\ln(H^2/(SK))}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}$:

$$C_{DI} = S\left(\tfrac{H}{S}\right)^{2\lambda} N(y) - Ke^{-rT}\left(\tfrac{H}{S}\right)^{2\lambda-2} N(y - \sigma\sqrt{T}), \qquad C_{DO} = C - C_{DI}.$$

The Monte Carlo uses 100,000 paths of 252 daily steps (seed 1) and sets the payoff to zero if any daily price is at or below H. Daily monitoring is less strict than continuous monitoring, so the simulated price is higher. The Broadie-Glasserman-Kou correction moves the barrier to $H\,e^{-0.5826\,\sigma\sqrt{\Delta t}}$ in the formula.

2.5 Delta hedging and strategies

The seller of a call receives the premium, buys Δ shares and rebalances at n equal dates. Cash grows at r. At expiry the final P&L is

$$\text{P\&L} = \text{cash}_T + \Delta_{n-1}S_T - \max(S_T - K, 0).$$

There are no costs or dividends. The test uses a 3-month at-the-money call, 2,000 simulated paths per case and seed 7.

For strategies, each leg is a (type, strike, quantity). The profit at expiry is $\sum_i q_i\,\text{payoff}_i(S_T) - \text{cost} \times e^{rT}$, where the cost is the sum of Black-Scholes prices. Break-evens are the sign changes on a grid of 1,601 prices between 60 and 140.

3 Results

Tree against Black-Scholes (ATM call, 1 year, Black-Scholes price 9.4134):

Steps	Tree price	Error
10	9.2179	−0.1956
100	9.3936	−0.0198
1000	9.4114	−0.0020

The error falls roughly in proportion to $1/n$ (ten times more steps, ten times smaller error).

American against European puts (500 steps). An American call without dividends has the same value as the European call (9.4094 for both).

Strike	European put	American put	Early exercise premium
80	0.8607	0.8830	0.0223
100	6.4540	6.7411	0.2871
120	19.2195	20.6068	1.3873
140	36.4932	40.0000	3.5068

At $K = 140$ the American put is worth its intrinsic value of 40: it is optimal to exercise at once.

Greeks. The formula and the finite difference agree to five decimals for the ATM call: delta 0.59871, gamma 0.01933, vega 0.38667, theta -0.01474 and rho 0.50457.

Barrier ($H = 90$): the vanilla call is 9.4134. The down-and-out call is 7.7402 by formula. Monte Carlo gives 8.0034 (95% interval ± 0.0869). The formula with the adjusted barrier gives 7.9700, which is inside the Monte Carlo interval.

Delta hedging (call sold for 4.3576):

Rebalances	Mean P&L	Std P&L
4	+0.0445	1.5724
13	+0.0112	0.9449
63	−0.0004	0.4349
252	+0.0023	0.2199

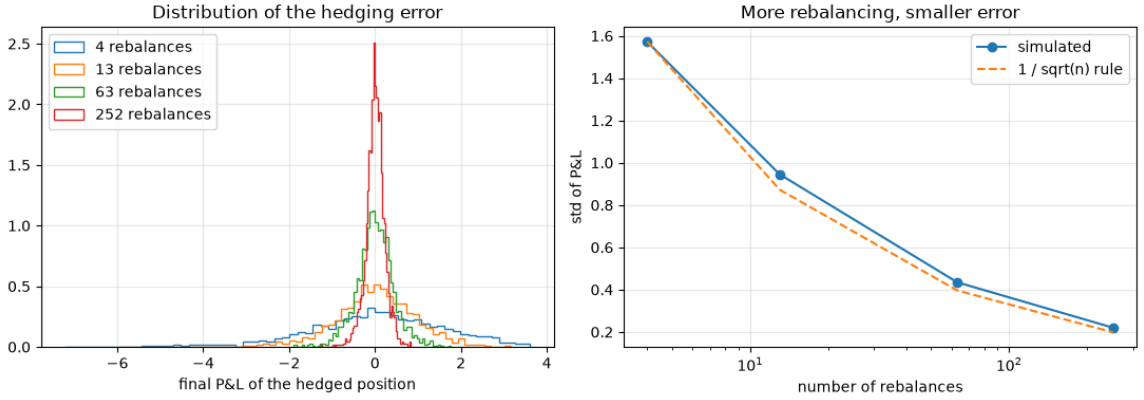


Figure 1: Distribution of the hedging error (left) and its standard deviation against the $1/\sqrt{n}$ rule (right).

Strategies (3 months, strikes in brackets; a negative cost is a credit):

Strategy	Cost	Max profit	Max loss	Break-even
Bull call spread (95/105)	5.07	4.89	−5.11	100.1
Long straddle (100)	7.97	unlimited	−8.03	92.0, 108.0
Long strangle (95/105)	3.98	unlimited	−4.01	91.0, 109.0
Butterfly (90/100/110)	3.66	6.31	−3.69	93.7, 106.3
Iron condor (85/90/110/115)	−1.07	1.08	−3.92	88.9, 111.1

4 Worked Example

Black-Scholes by hand for $S = K = 100$, $T = 1$, $r = 3\%$, $\sigma = 20\%$:

$$d_1 = \frac{0 + (0.03 + 0.02) \times 1}{0.20} = 0.25, \qquad d_2 = 0.25 - 0.20 = 0.05,$$
$$N(d_1) = 0.59871, \qquad N(d_2) = 0.51994, \qquad Ke^{-rT} = 97.0446,$$
$$C = 100 \times 0.59871 - 97.0446 \times 0.51994 = 59.871 - 50.457 = 9.4134.$$

The put is $97.0446 \times 0.48006 - 100 \times 0.40129 = 6.4580$. Parity: $9.4134 - 6.4580 = 2.9554 = 100 - 97.0446$. The numbers $N(d_1) = 0.59871$ and $N(d_2) = 0.51994$ also give the delta and, through $KTe^{-rT}N(d_2)/100$, the rho of 0.50457 in the Greeks results above.

Barrier correction. For daily monitoring, $\sigma\sqrt{\Delta t} = 0.20 \times \sqrt{1/252} = 0.01260$, so

$$H_{adj} = 90 \times e^{-0.5826 \times 0.01260} = 90 \times e^{-0.00734} = 89.34.$$

Lowering the barrier makes the option less likely to be knocked out, so the price rises from 7.7402 to 7.9700.

The $1/\sqrt{n}$ rule. From 4 rebalances (std 1.5724), the rule predicts $1.5724 \times \sqrt{4/252} = 0.198$ at 252 rebalances. The simulation gives 0.2199, about 11% higher.

5 Limits

- **Textbook assumptions:** constant volatility and rate, no dividends, no transaction costs, and continuous trading in the limit. Real markets have a smile and costs.
- **Small samples:** the hedging test uses 2,000 paths per case and one seed. The mean P&L values of 0.0445 against 0.0023 are noise of the order of the standard error (about 0.035 for 4 rebalances).
- **Barrier monitoring:** the Monte Carlo with 252 steps sits 0.033 above the corrected formula. The correction is an approximation.
- **No market data:** all prices are model prices, so the notebooks check internal consistency, not agreement with traded prices. These are model outputs, not recommendations.